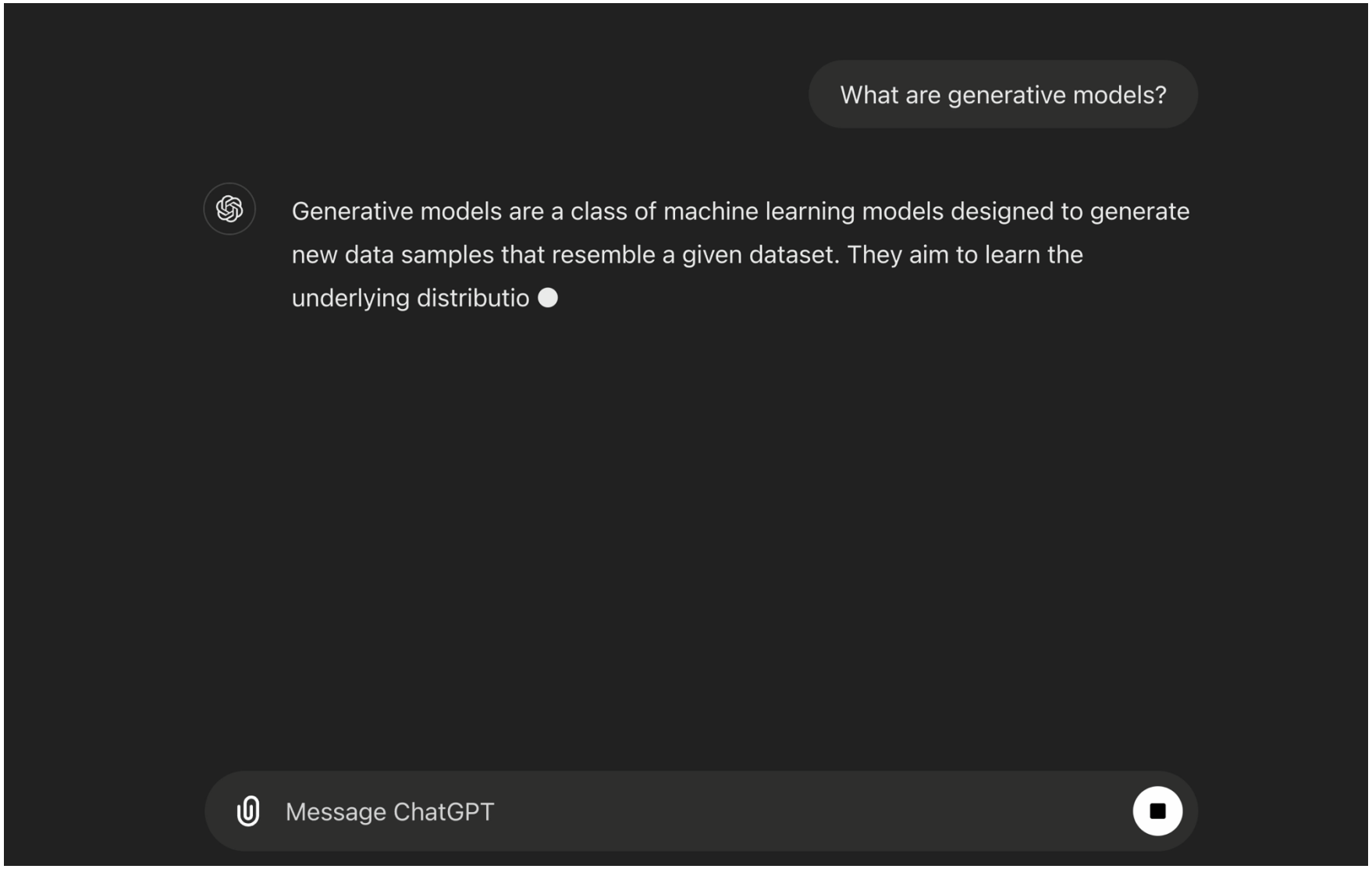
Lecture 13

# Generative Models: Introduction

Speaker: Kaiming He

# The “GenAI” Era

## Chatbot and natural language conversation



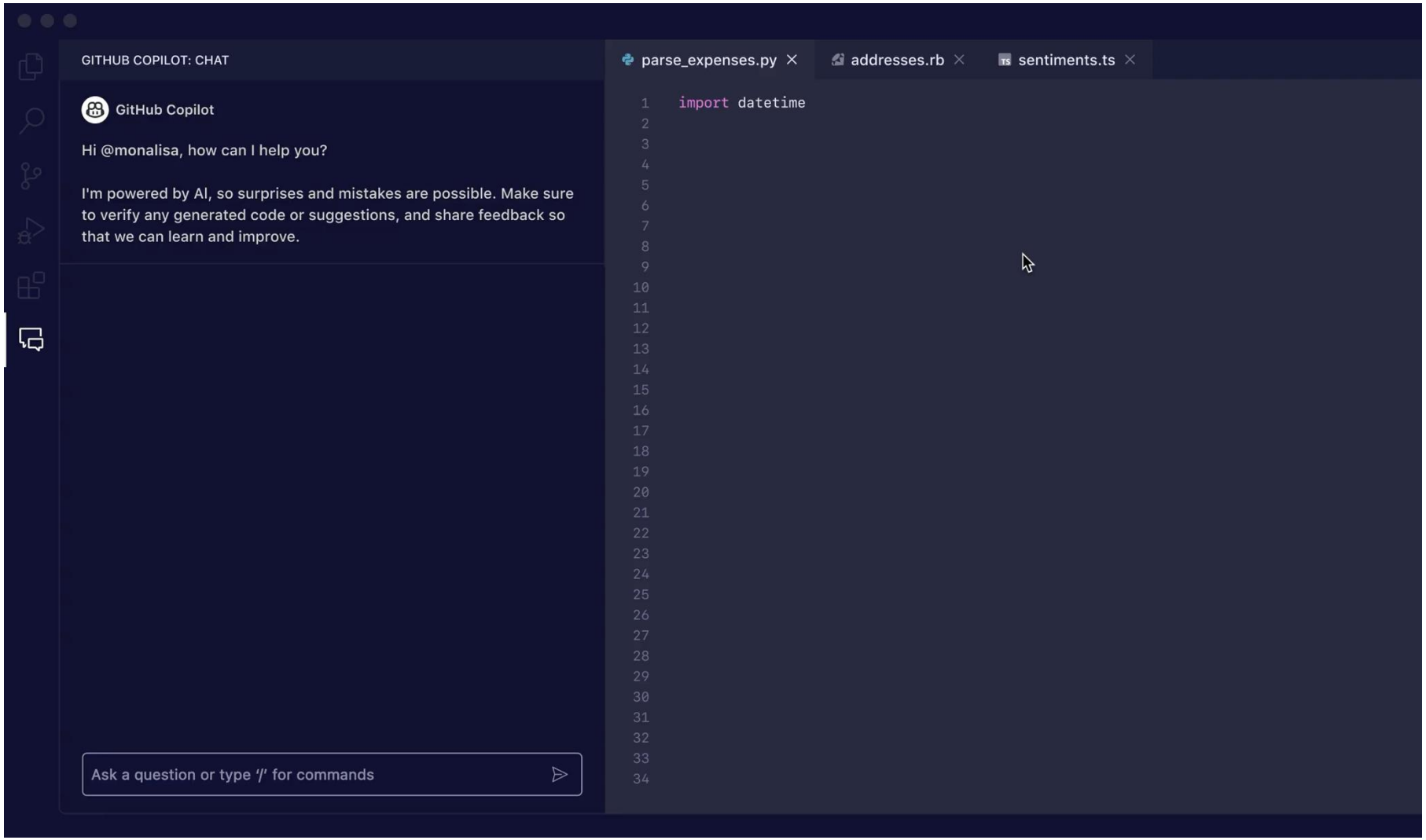
# The “GenAI” Era

Text-to-image generation

*Generated by Stable Diffusion 3 Medium.*
*Prompt: teddy bear teaching a course, with "generative models" written on blackboard*

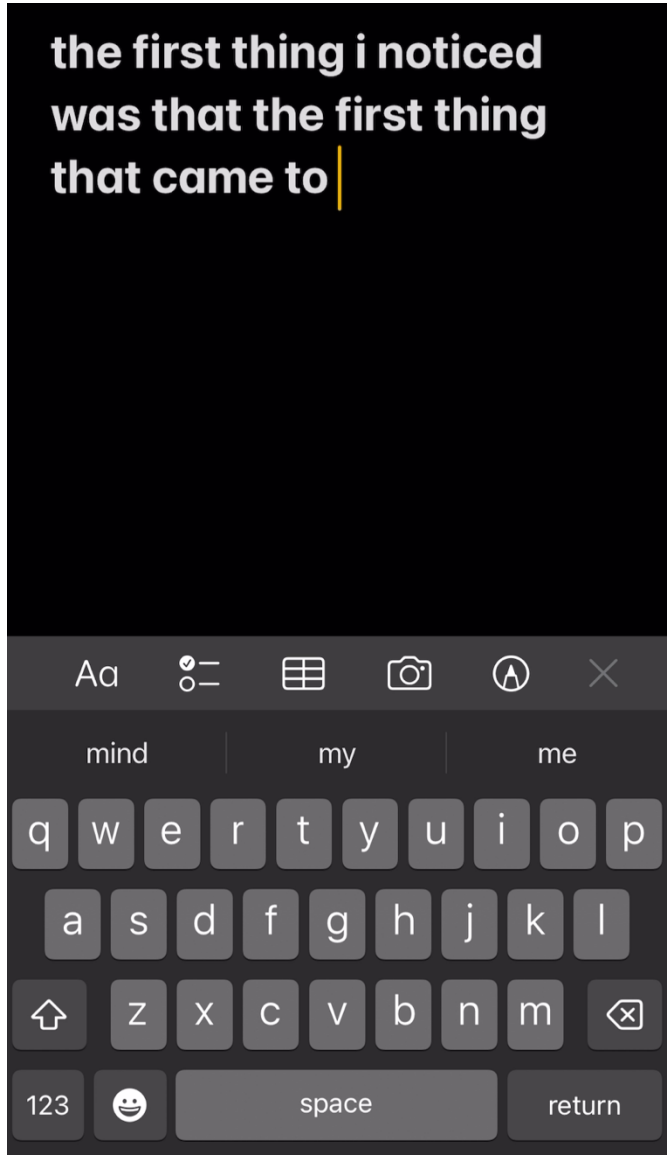
# The “GenAI” Era

## AI assistant for code generation



# Generative Models before the “GenAI” Era

Your keyboard



# Generative Models before the “GenAI” Era

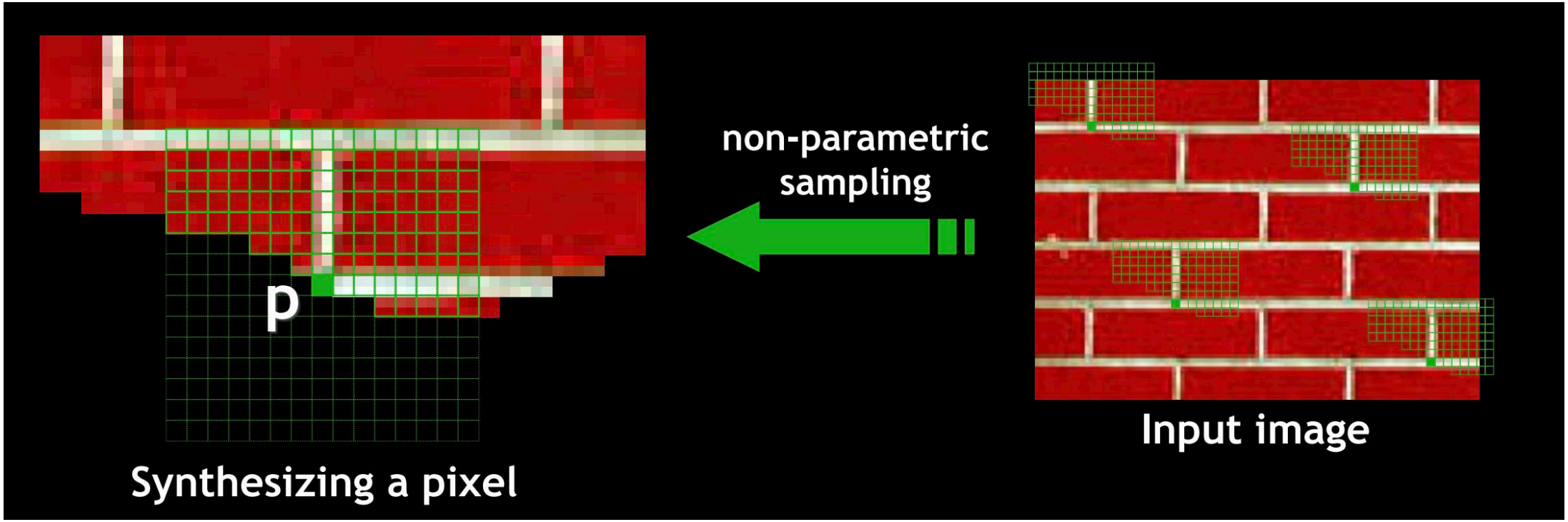
## **2009, PatchMatch:** Photoshop’s Content-aware Fill

PatchMatch: A Randomized Correspondence Algorithm for Structural Image Editing, SIGGRAPH 2009

# Generative Models before the “GenAI” Era

**1999, the Efros-Leung algorithm** for texture synthesis

In today’s word: this is an **Autoregressive** model

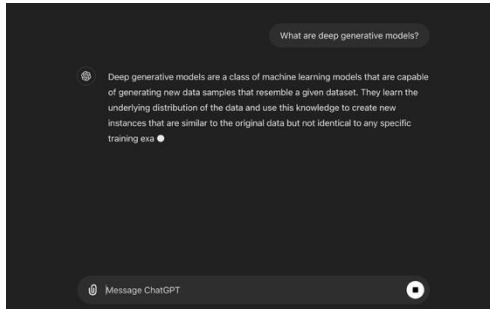



Texture Synthesis by Non-parametric Sampling, ICCV 1999

# What are Generative Models?

# What do these scenarios have in common?

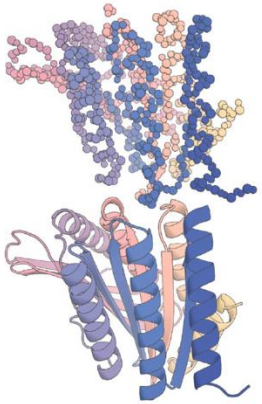
- There are **multiple** predictions to one input.
- Some predictions are more **plausible**.
- Training data may contain **no solution**.
- Output may be **more complex**, more informative, and higher-dimensional




Chatbot

Image generation

Video generation



Protein generation

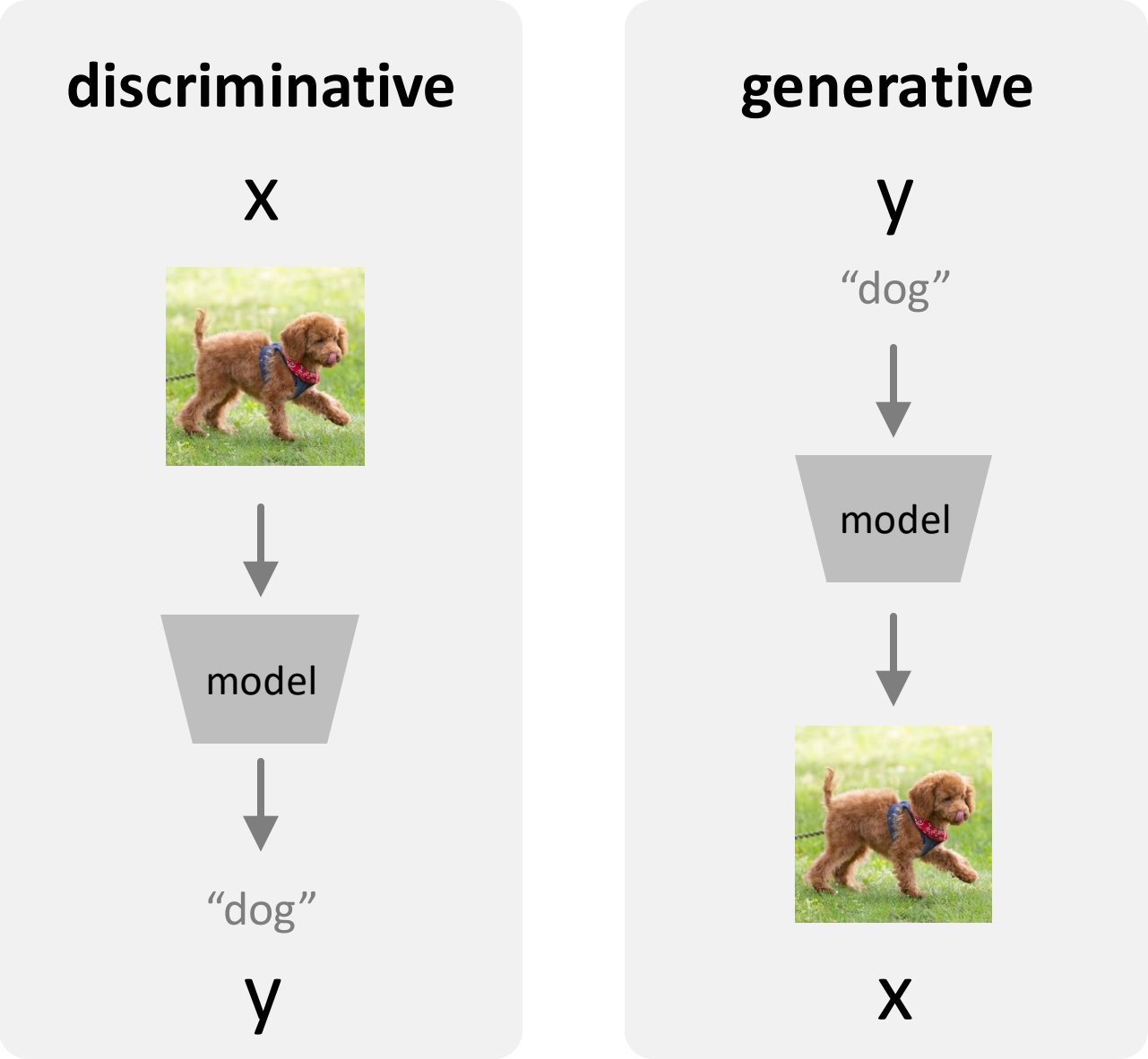
# Discriminative Models vs. Generative Models

**Discriminative Model**

- “sample” x ⇨ “label” y
- one desired output

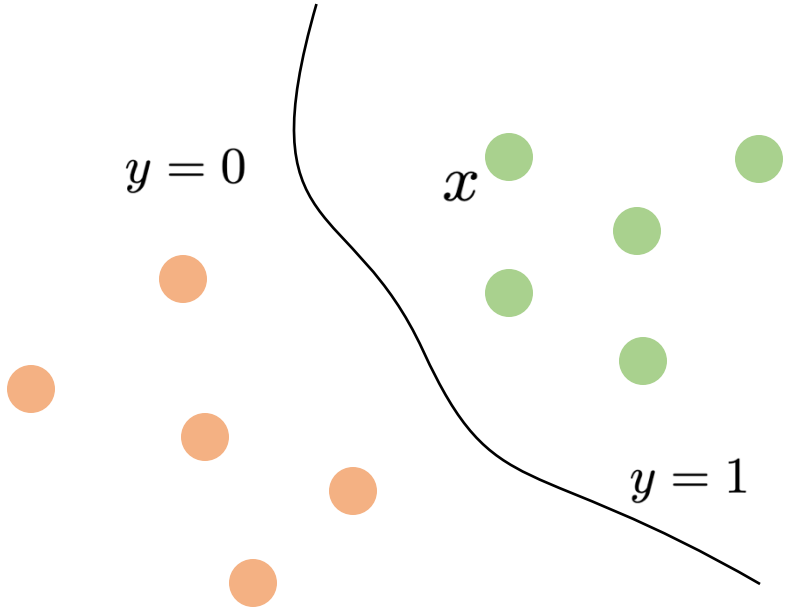
**Generative Model**

- “label” y ⇨ “sample” x
- many possible outputs

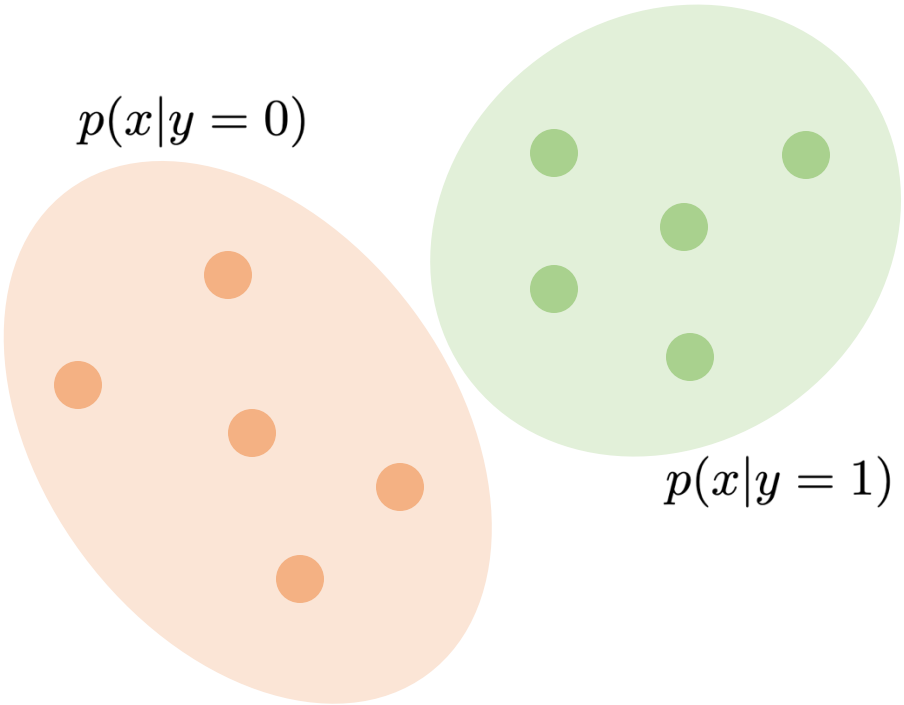


# Discriminative Models vs. Generative Models

**discriminative** $p(y|x)$




**generative** $p(x|y)$




- Generative models can be discriminative: Bayes' rule
- Can discriminative models be generative?

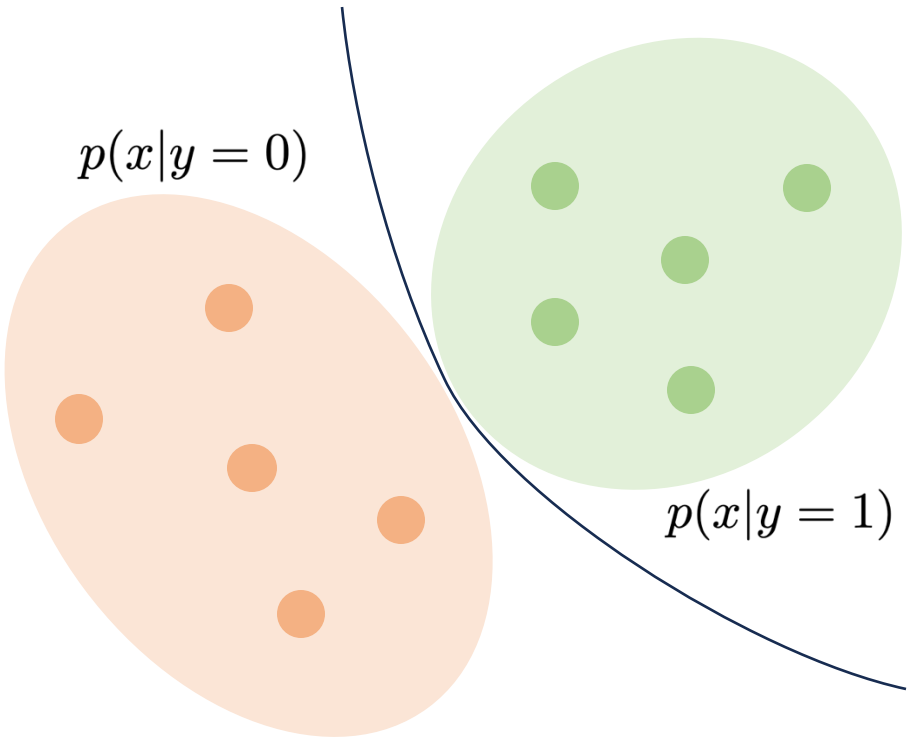
- Generative models can be discriminative: Bayes' rule

$$p(y|x) = p(x|y)\frac{p(y)}{p(x)}$$

discriminative generative

assuming known prior (uniform categories)

constant for given x




decision boundary:

$$p(x \mid y=0) = p(x \mid y=1)$$

- Generative models can be discriminative: Bayes' rule

$$p(y|x) = p(x|y)\frac{p(y)}{p(x)}$$

discriminative generative

assuming known prior (uniform categories)

constant for given x

- Can discriminative models be generative?

$$p(x|y) = p(y|x)\frac{p(x)}{p(y)}$$

generative discriminative

**unknown prior distribution of** x ("natural" images?)

constant for given y

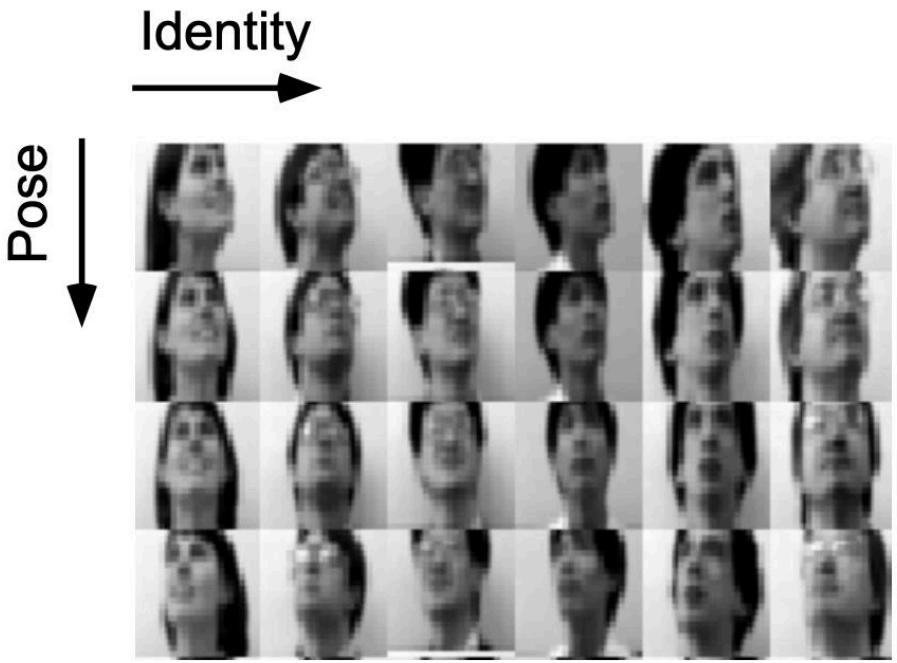
- The challenge is on **probabilistic modeling**

# Probabilistic Modeling

- Where does probability come from?
- Assuming underlying **distributions of data generation process**

**example**:

- latent factors z (pose, lighting, scale, ...)
- z has simple distributions
- observations x are rendered by a graphics model, (a function on z )
- x has complex distributions




- Probability itself is a modeling assumption

Figure from: W. T. Freeman, J. B. Tenenbaum, “Learning Bilinear Models for Two-Factor Problems in Vision”, 1996

# Probability itself is a modeling assumption

- There may be **no true "underlying" distribution** behind the data.
- Even if it exists, we can only observe **a finite sample set** of it.
- Models must **extrapolate** from observations to represent a distribution.
- Leads to **overfitting vs. underfitting** (just as in discriminative models)

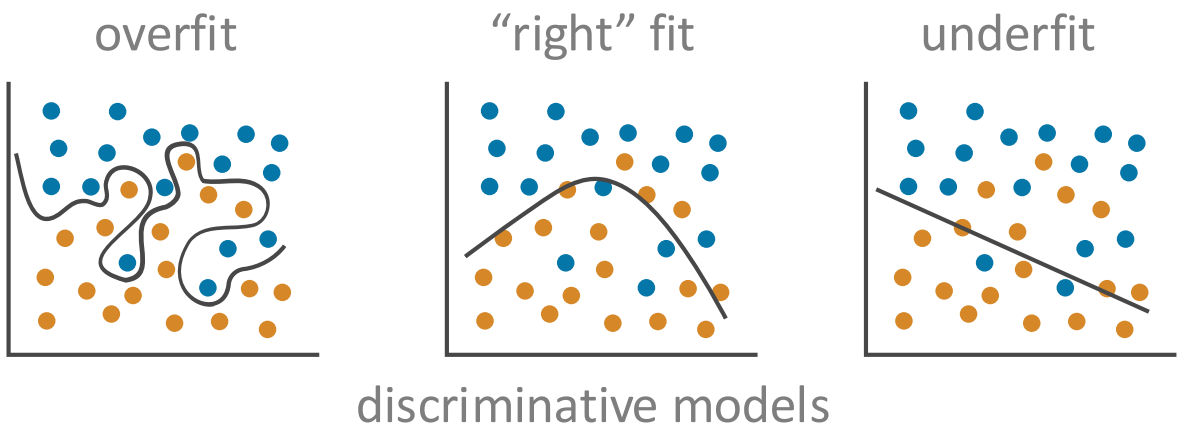
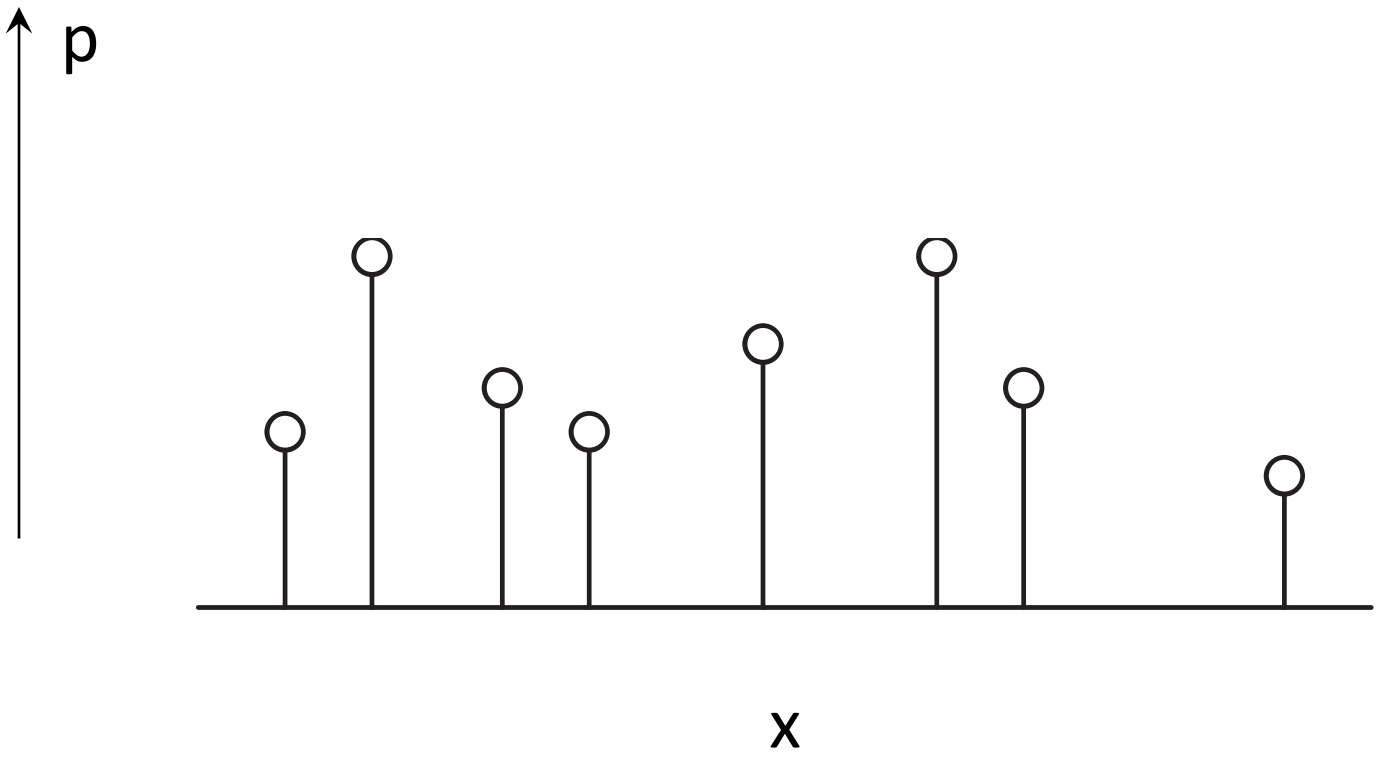



Figure credit: https://www.mathworks.com/discovery/overfitting.html

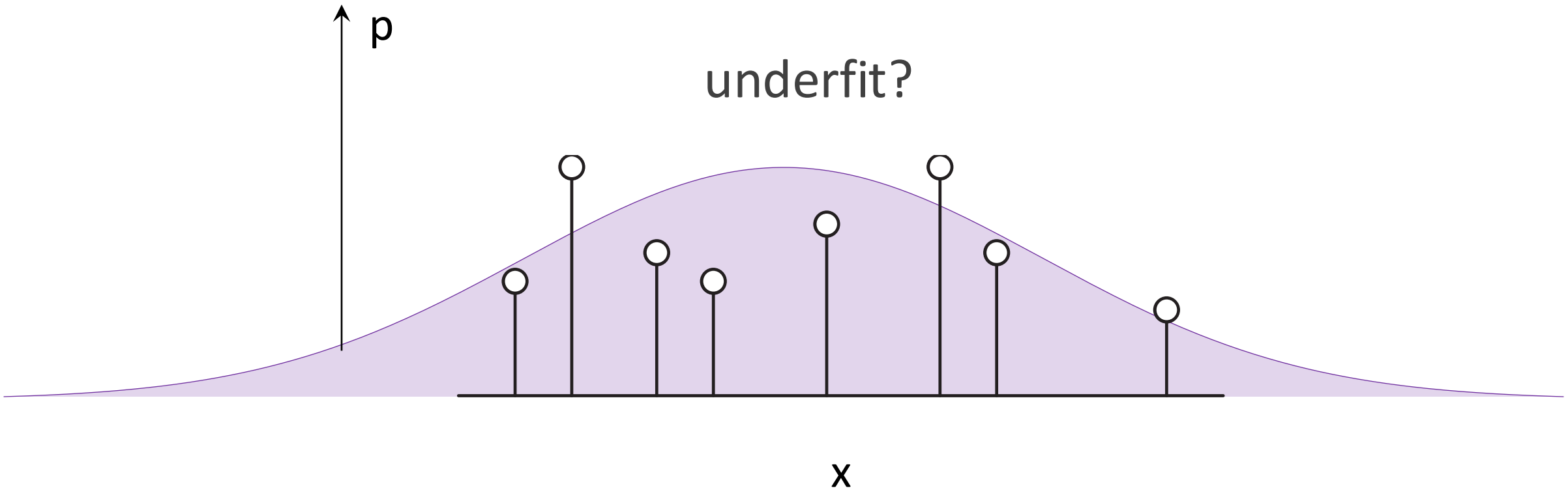
# Probability itself is a modeling assumption

- We can only observe **a finite sample set** of an unknown distribution.



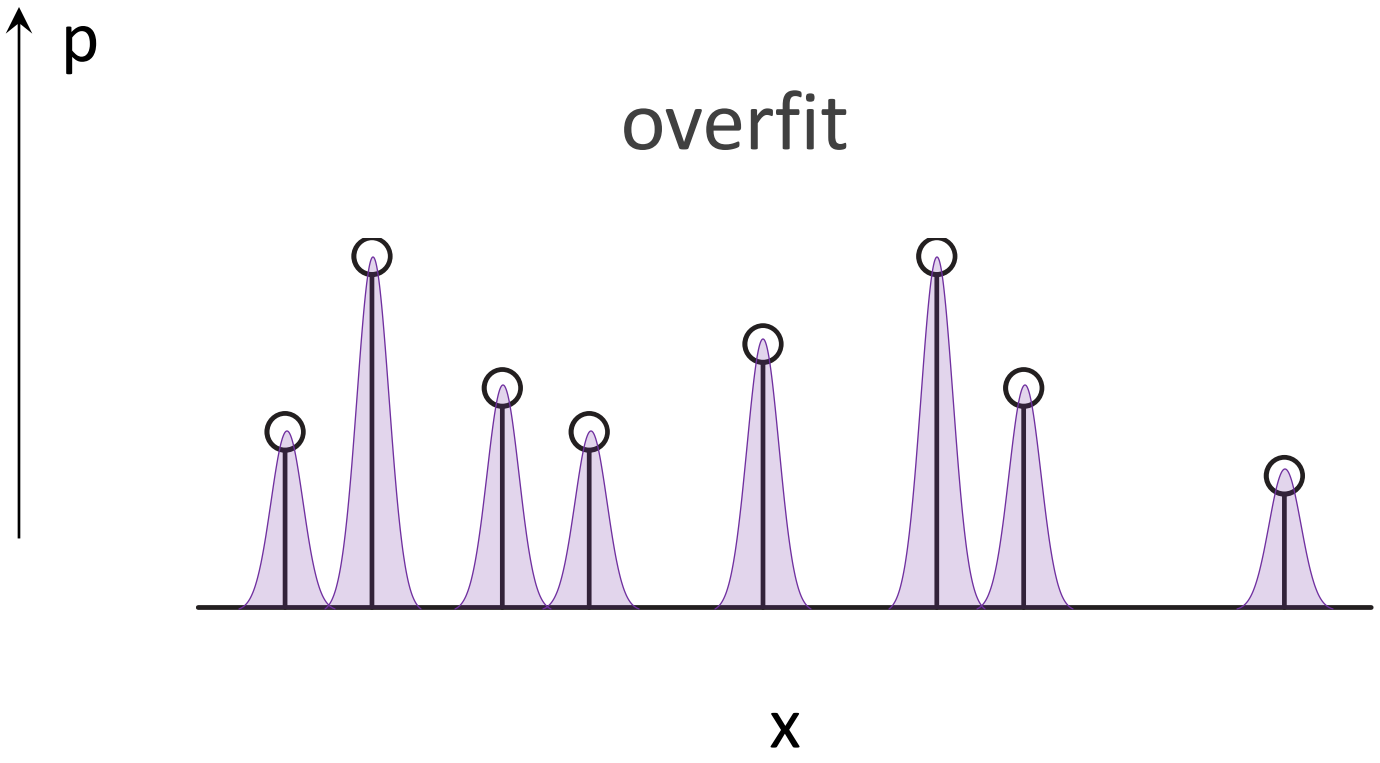
# Probability itself is a modeling assumption

- **Underfitting**: the probabilistic model is too simple.



# Probability itself is a modeling assumption

- **Overfitting**: the probabilistic model does not extrapolate/generalize



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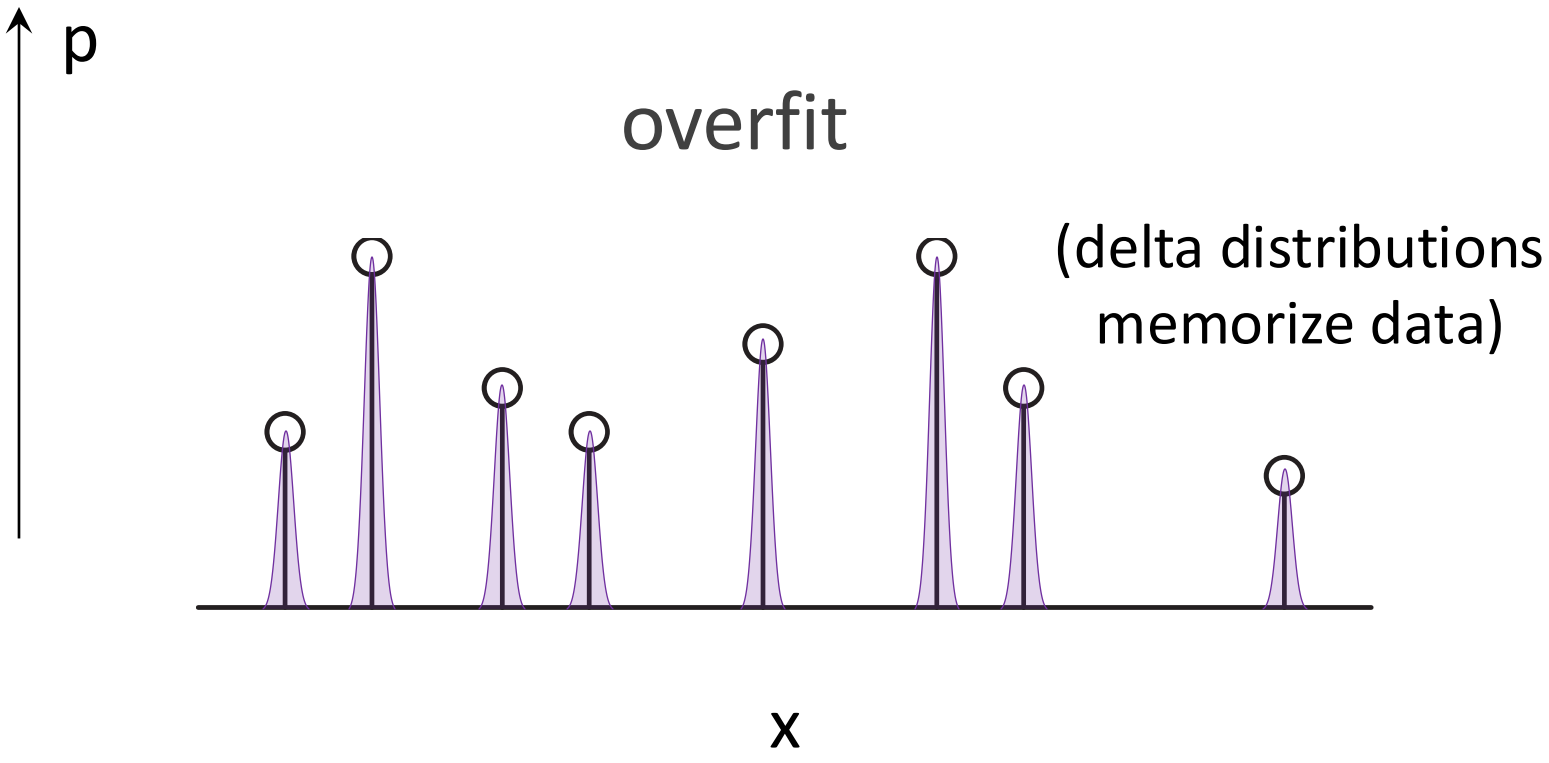


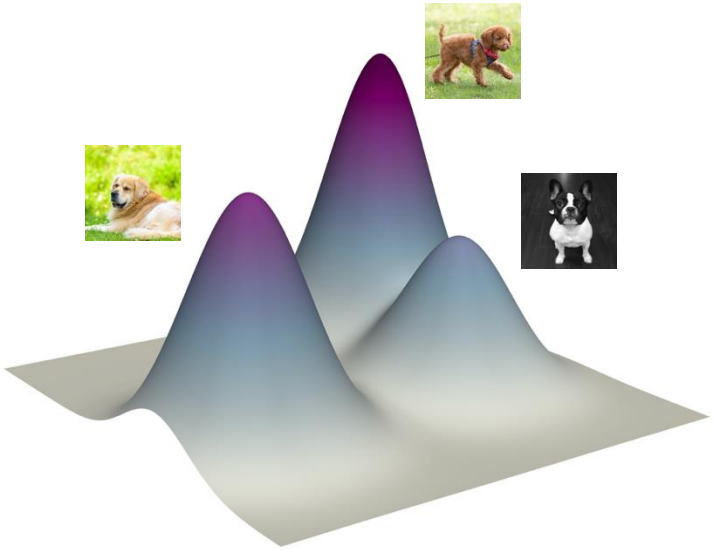
# Generative Modeling is Probabilistic Modeling

data

# Generative Modeling is Probabilistic Modeling

data

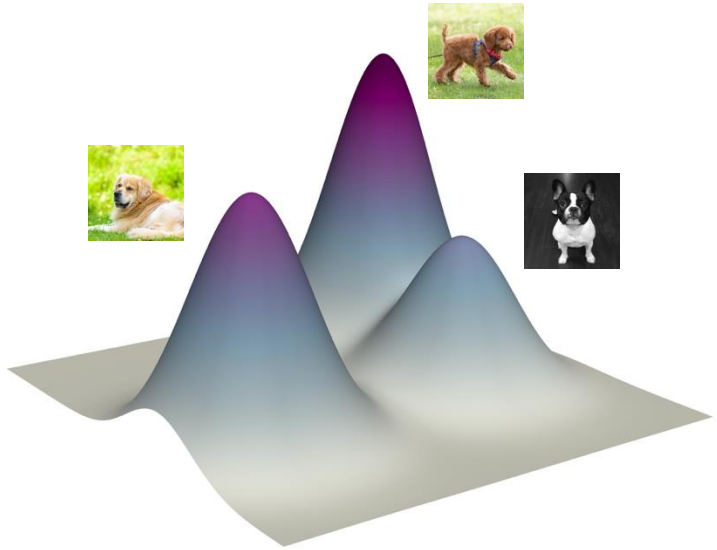
data distribution



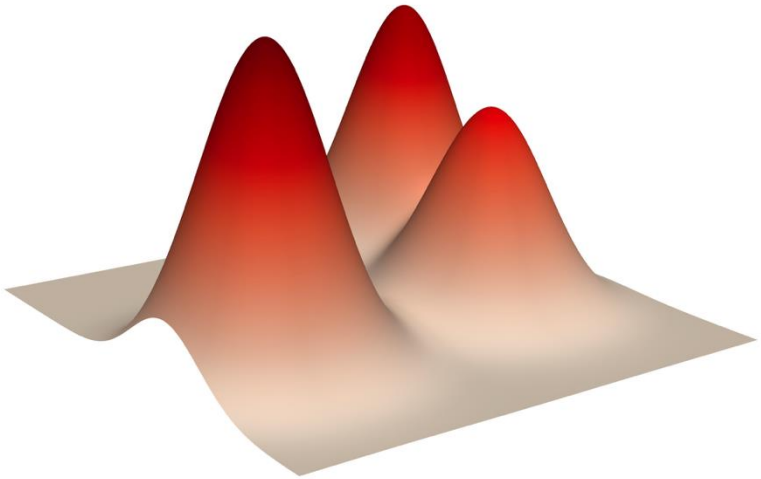
This in itself is part of modeling

# Generative Modeling is Probabilistic Modeling

data

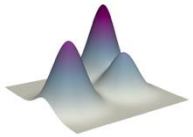
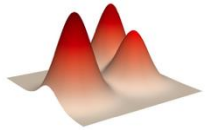


data distribution

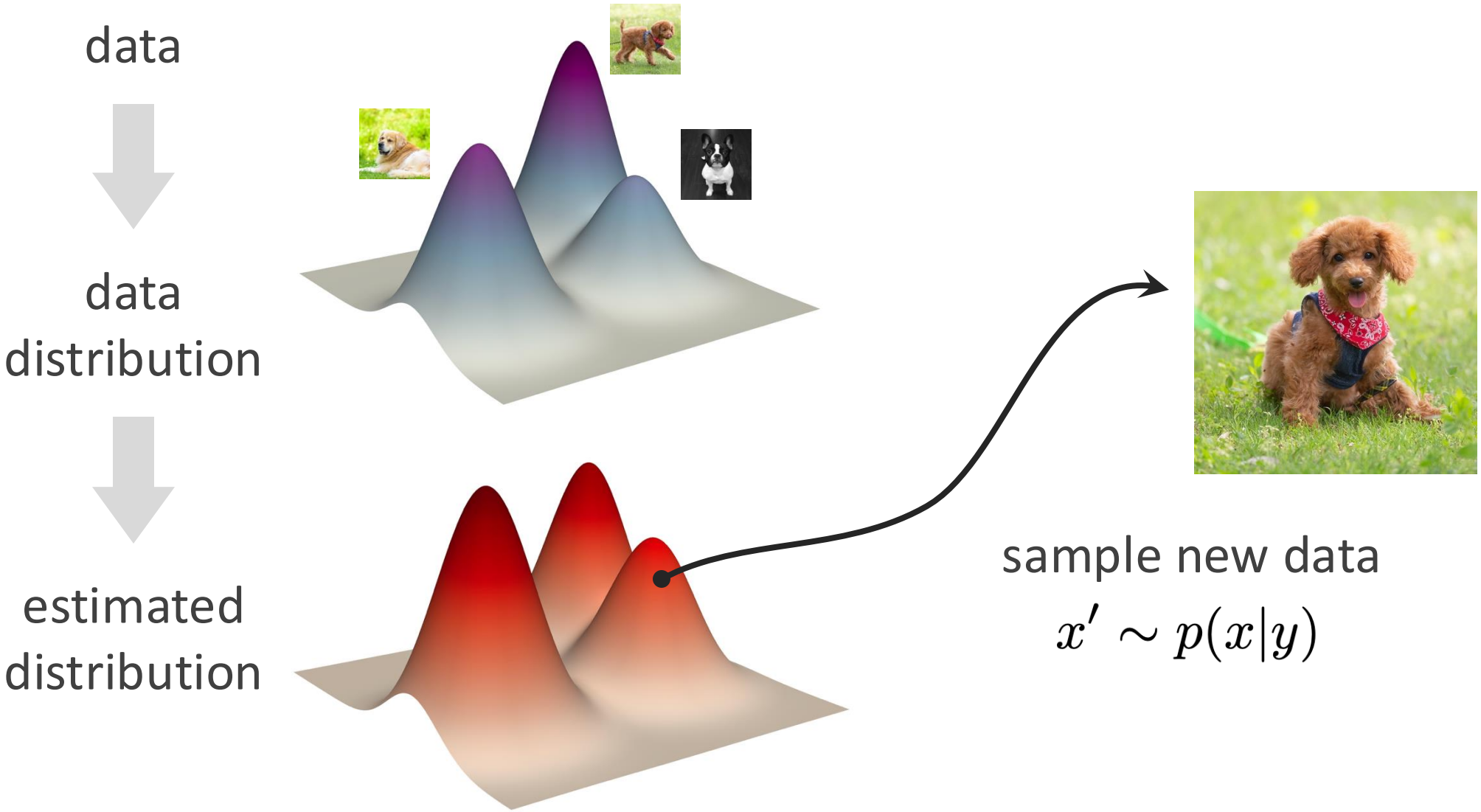


estimated distribution

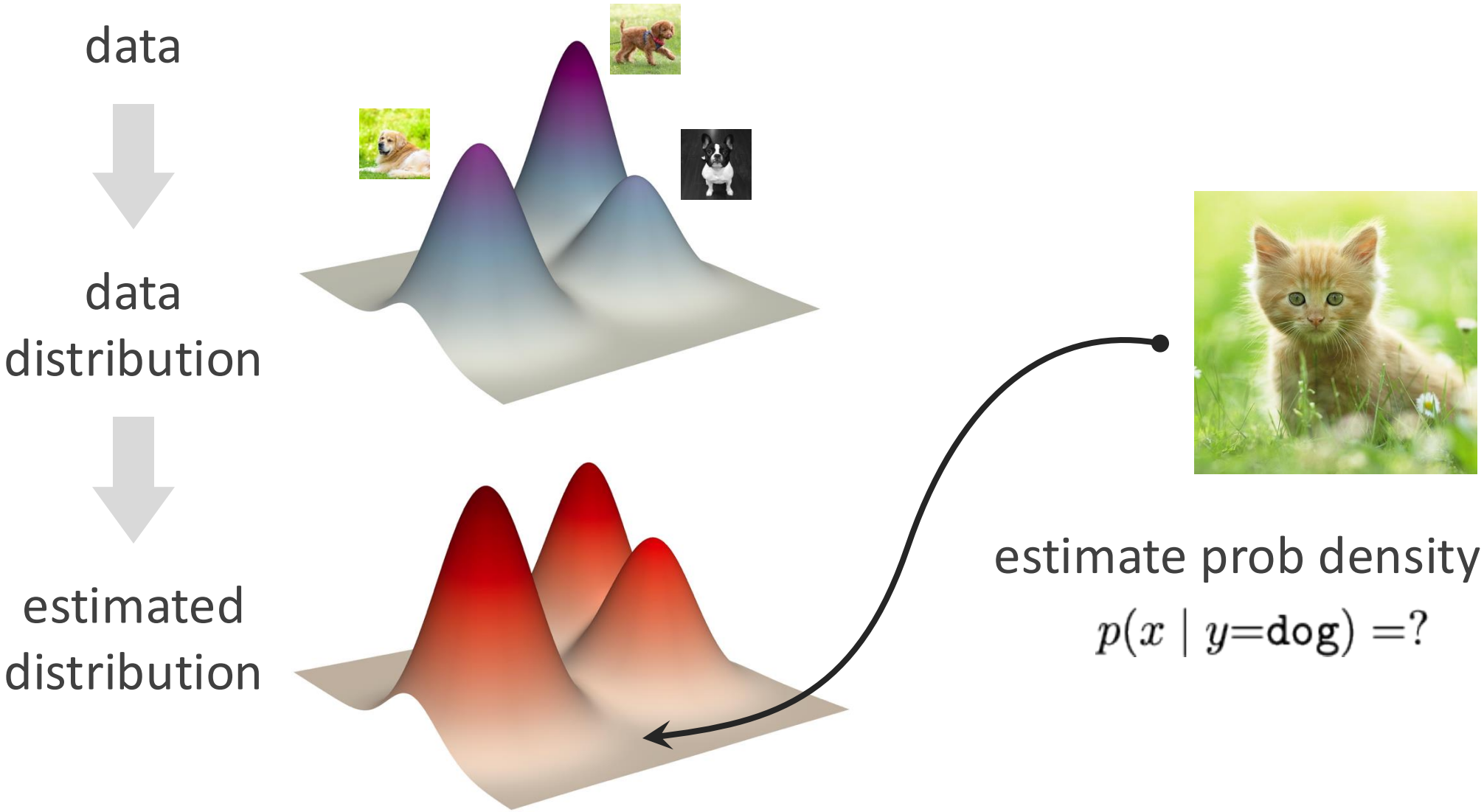
Optimize a loss function

$\mathcal{L}($  ,  $)$

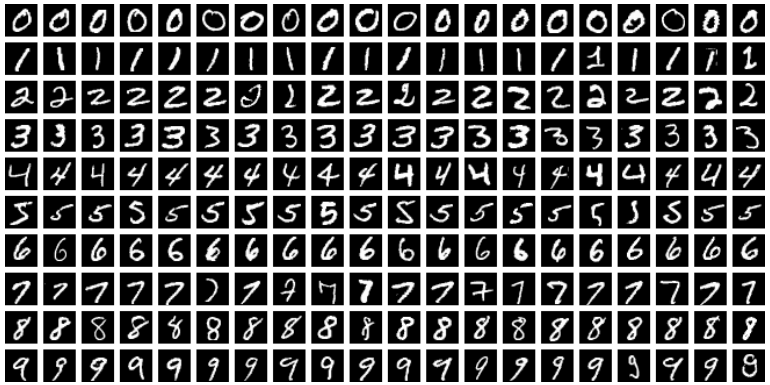
# Generative Modeling is Probabilistic Modeling



# Generative Modeling is Probabilistic Modeling



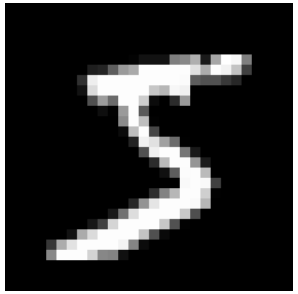
# Case study: a minimalist generative model?



data set

random index

retrieve



- Is this a generative model? Yes
- Is this a “good” generative model? No - not generalize?

# Generative Modeling is Probabilistic Modeling

**Notes:**

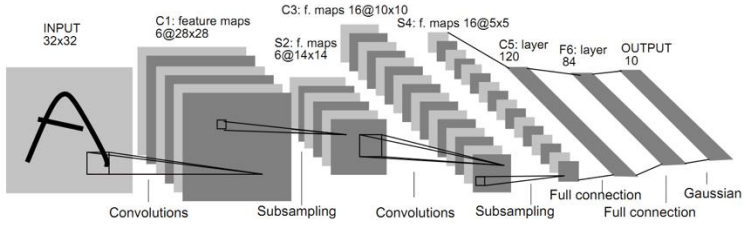
- Generative models involve probabilistic models hypothesized by humans.
- Probabilistic modeling is not just the work of neural nets (even in the context of deep learning).
- Probabilistic modeling is a popular way, but not the only way.
- "*All models are wrong, but some are useful*."*

* Quote George Box

# Deep Generative Models

- Deep learning is **representation learning**
- **Discriminative**: represent data **instances**
  - one data point: $x \to f_\theta(x)$
  - loss w.r.t. one label: $\mathcal{L}(y, f_\theta(x))$

$$x \longrightarrow f(x)$$



# Deep Generative Models

- Deep learning is **representation learning**

$x \longrightarrow f(x)$

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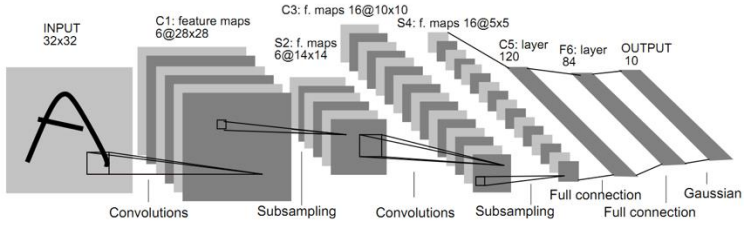
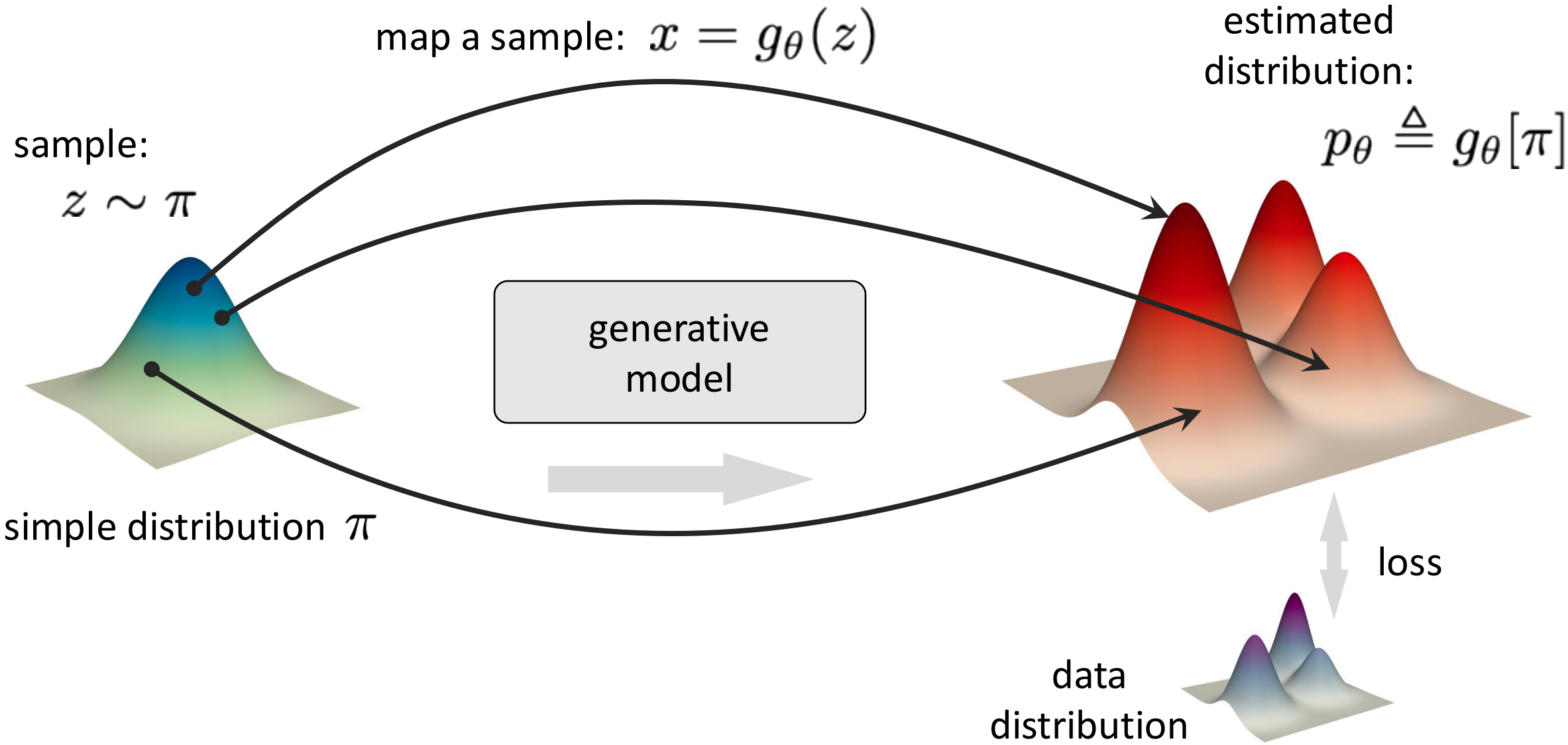



$g_\theta[\pi] \longleftarrow \pi$

- **Generative**: represent data **distribution**
  - distribution-to-distribution mapping: $\pi \rightarrow g_\theta[\pi]$ (example: $\pi$ is Gaussian)
  - loss w.r.t. true (but known) distribution: $\mathcal{L}(p_{\text{data}}, g_\theta[\pi])$

# Learning to Represent Distributions

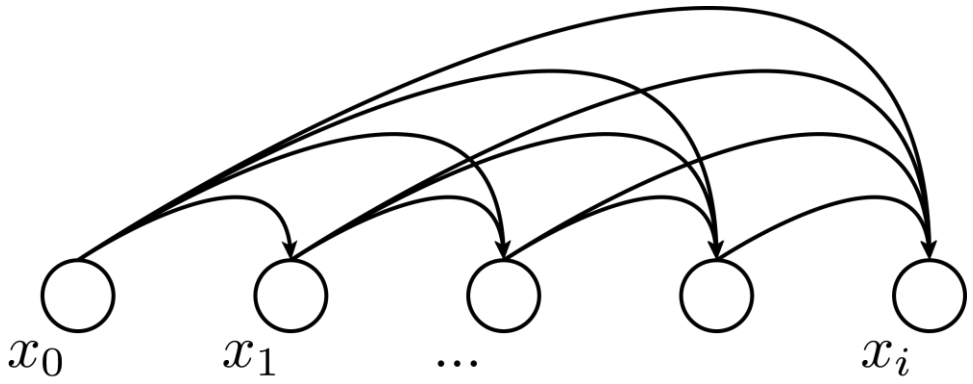


# Learning to Represent Distributions

- Not all parts of a generative model are learned.

Case study:
Autoregressive model

This dependency structure is designed (not learned).

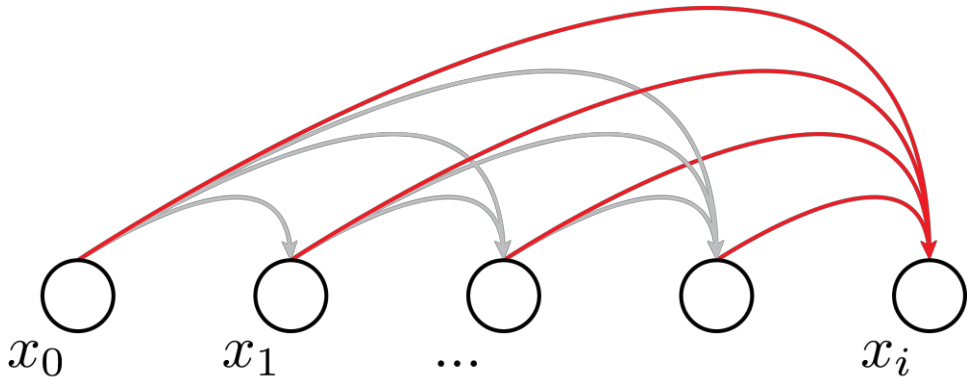


# Learning to Represent Distributions

- Not all parts of a generative model are learned.

Case study:
Autoregressive model

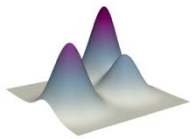
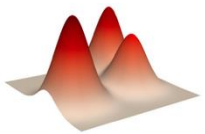
This probability is learned




$$p(x_i \mid x_0, x_1, \ldots)$$

# Objective Functions in Generative Models

- The objective is a *functional*: it operates on functions

$\mathcal{L}($  $,$  $)$

a functional

a function

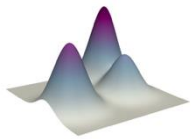
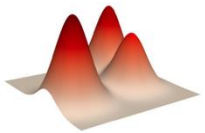
$p_{\text{data}}(x)$

another function

$p_\theta(x)$

# Objective Functions in Generative Models

- Case study: Kullback–Leibler (KL) Divergence

$\mathcal{L}($  ,  $)$

x is a dummy variable

$$\mathcal{D}_{\mathrm{KL}}(p_{\mathrm{data}} \,||\, p_\theta) \triangleq \mathbb{E}_{x \sim p_{\mathrm{data}}}\left[\log \frac{p_{\mathrm{data}}(x)}{p_\theta(x)}\right]$$

$\mathcal{D}$ 's inputs: $p_\theta, p_{\mathrm{data}}$

# What makes the objectives difficult to design?

- Analytical formulations
  - Only for limited families (Gaussian, uniform, categorical, ...)
  - Low-dimensional
  - Example: Gaussian => L2 loss

- Monte Carlo estimation
  - Distributions unknown in analytical form
  - We can only sample from them
  - Example: $x, y \sim p_{\text{data}}(x, y)$

- Divergence measures
  - often not directly computable (a surrogate is used)
  - Example: GAN

See also "What Makes Modeling Information Difficult?" in Lecture 10

# Elements of Deep Generative Models

- **Formulation:** a real-world problem => probabilistic modeling
  - designed structures
  - learnable components
- **Representation**: use deep neural nets to represent distributions
- **Objectives**: discrepancy between distributions
- **Optimization**: differentiable, computable, and tractable
- **Inference**:
  - sampler: able to produce new samples
  - density estimator: (optional) able to evaluate $p(x)$

# Generative Modeling of Real-world Problems

# Generative Modeling of Real-world Problems

- Generative models are about $p(x|y)$

**What can be y?**

- condition
- constraint
- labels
- attributes

- more abstract
- less informative

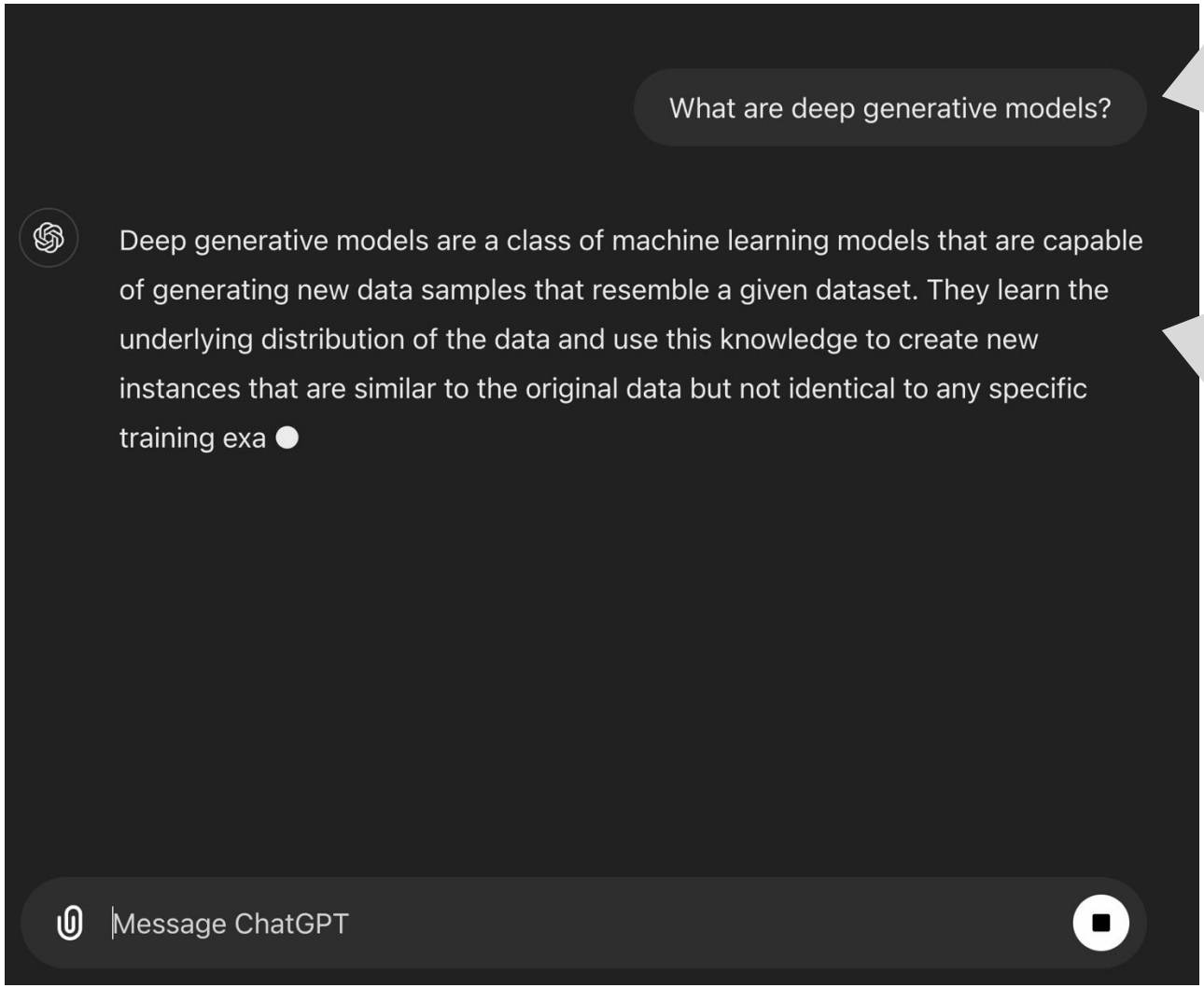
**What can be x?**

- "data"
- samples
- observations
- measurements

- more concrete
- more informative

# Generative Modeling as $p(x|y)$: Case Study

- **Natural language conversation**




y: prompt

x: response of the chatbot

# Generative Modeling as $p(x|y)$: Case Study

- **Text-to-image/video generation**

*Prompt: teddy bear teaching a course, with "generative models" written on blackboard*

y: text prompt

x: generated visual content

Image generated by Stable Diffusion 3 Medium

# Generative Modeling as p(x|y): Case Study

- **Class-conditional image generation**

*"red fox"*

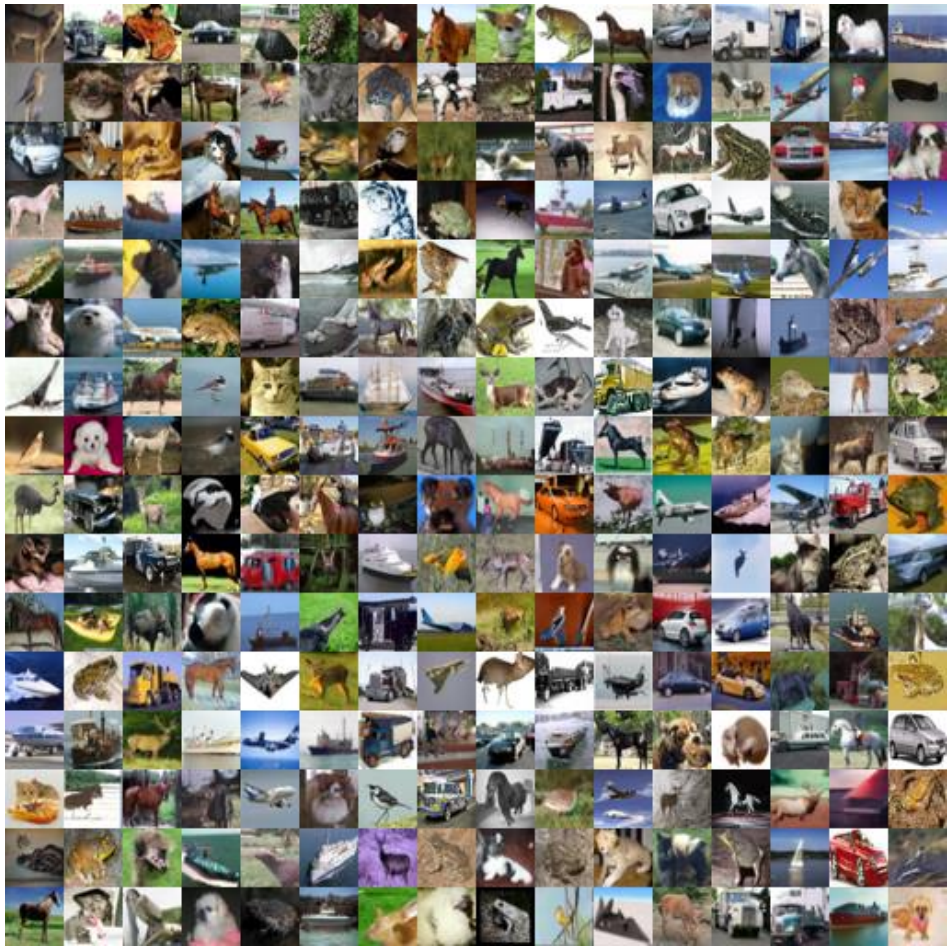
y: class label

x: generated image

Image generated by: Li, et al. Autoregressive Image Generation without Vector Quantization, 2024

# Generative Modeling as p(x|y): Case Study

- **"Unconditional" image generation**



y: an implicit condition

*"images following CIFAR10 distribution"*

x: generated CIFAR10-like images

- p(x|y): images ~ CIFAR10
- p(x): all images

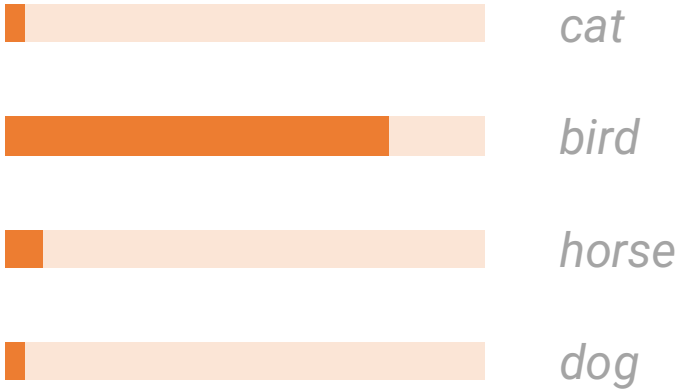
Images generated by: Karras, et al. Elucidating the Design Space of Diffusion-Based Generative Models, NeurIPS 2022

# Generative Modeling as p(x|y): Case Study

- **Classification** (often not viewed as generative)

y: an image as the “condition”

x: probability of classes conditioned on the image

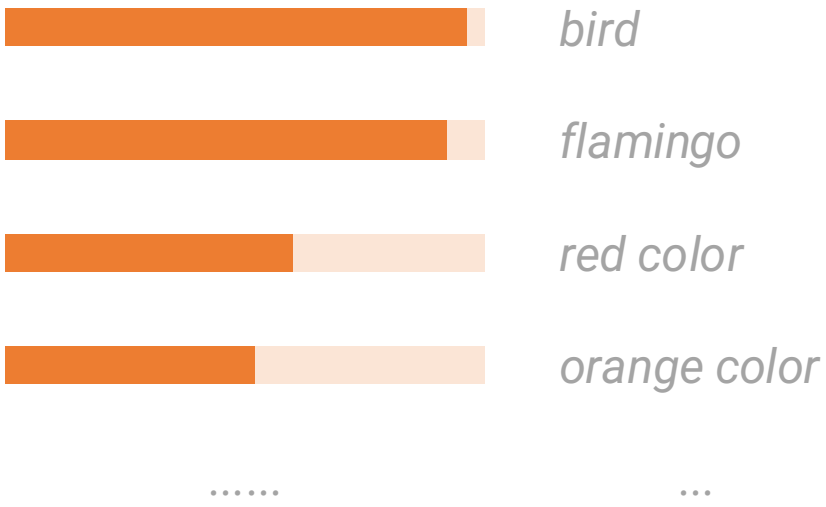


# Generative Modeling as p(x|y): Case Study

- **Open-vocabulary recognition**

y: an image as the “condition”

x: plausible descriptions conditioned on the image



# Generative Modeling as p(x|y): Case Study

- **Image captioning**

y: an image as the “condition”

x: plausible descriptions conditioned on the image

```
a baseball player with a catcher and umpire on top of a baseball field.
a baseball player is sliding into a base.
a baseball player swings at a pitch with the pitcher and umpire behind him.
baseball player with bat in the baseball game.
a batter in the process on the bat in a baseball game.
```

figure credit: https://github.com/GoogleCloudPlatform/asl-ml-immersion/blob/master/notebooks/multi_modal/solutions/image_captioning.ipynb

# Generative Modeling as p(x|y): Case Study

- **Visual dialogue**

User What is unusual about this image?

Source: https://www.barnorama.com/wp-content/uploads/2016/12/03-Confusing-Pictures.jpg

GPT-4 The unusual thing about this image is that a man is ironing clothes on an ironing board attached to the roof of a moving taxi.

y: image and text prompt

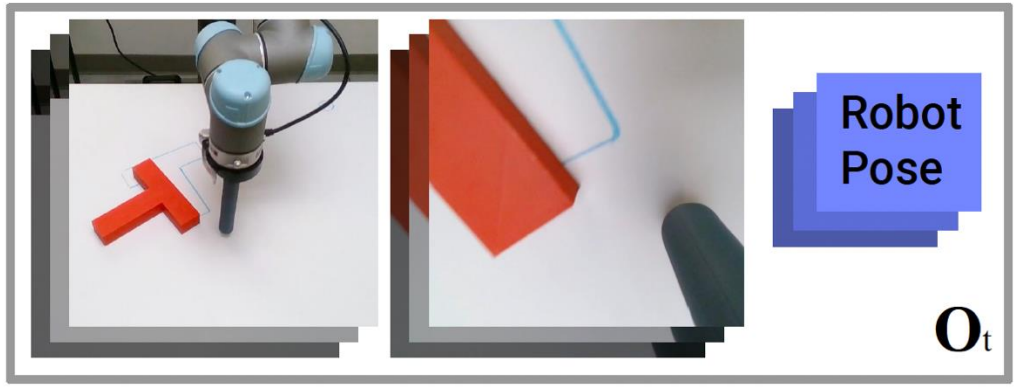
x: response of the chatbot

Figure from: GPT-4 Technical Report, 2023

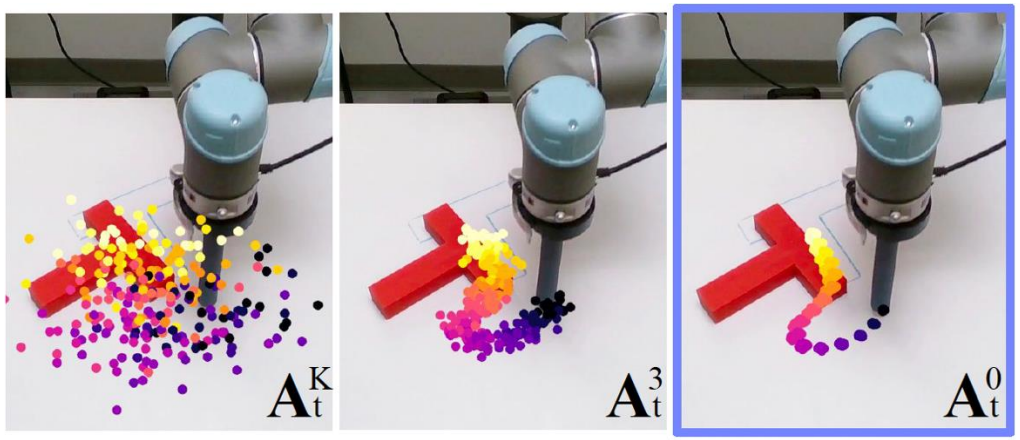
# Generative Modeling as p(x|y): Case Study

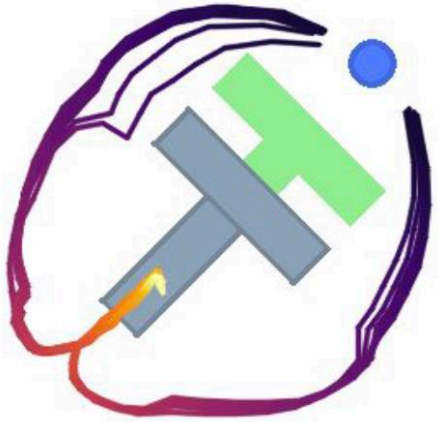
- **Policy Learning in Robotics**

y: visual and other sensory observations




x: policies (probability of actions)



Chi, et al. Diffusion Policy: Visuomotor Policy Learning via Action Diffusion, RSS 2023

# Generative Modeling of Real-world Problems

- Generative models are about $p(x|y)$

**What can be y?**

- ~~condition~~
- ~~constraint~~
- ~~labels~~
- ~~attributes~~

- ~~more abstract~~
- ~~less informative~~

**What can be x?**

- ~~"data"~~
- ~~samples~~
- ~~observations~~
- ~~measurements~~

- ~~more concrete~~
- ~~more informative~~

# Generative Modeling of Real-world Problems

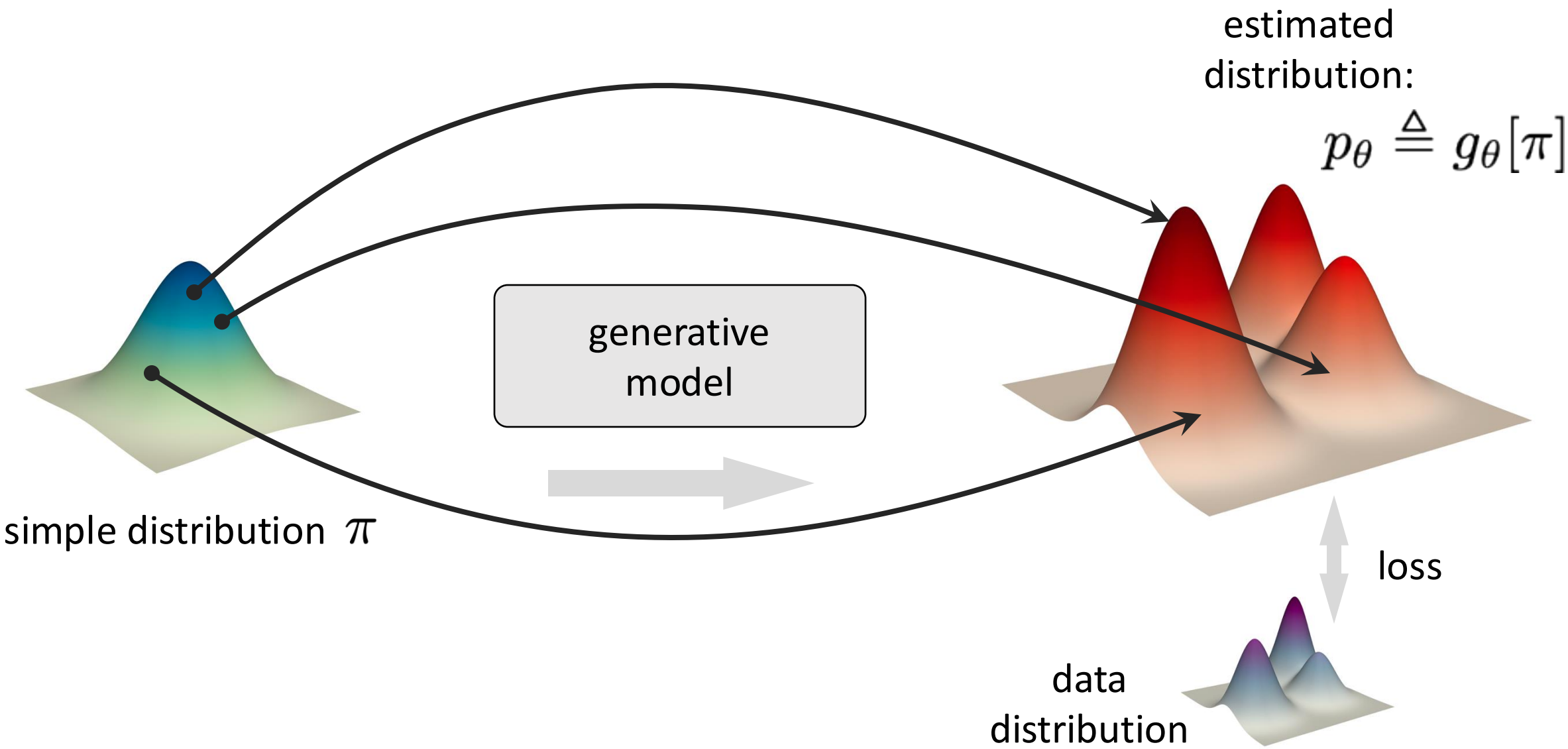
- Generative models are about $p(x|y)$

- There is no conceptual obstacle to formulating any real-world problem from a generative perspective.

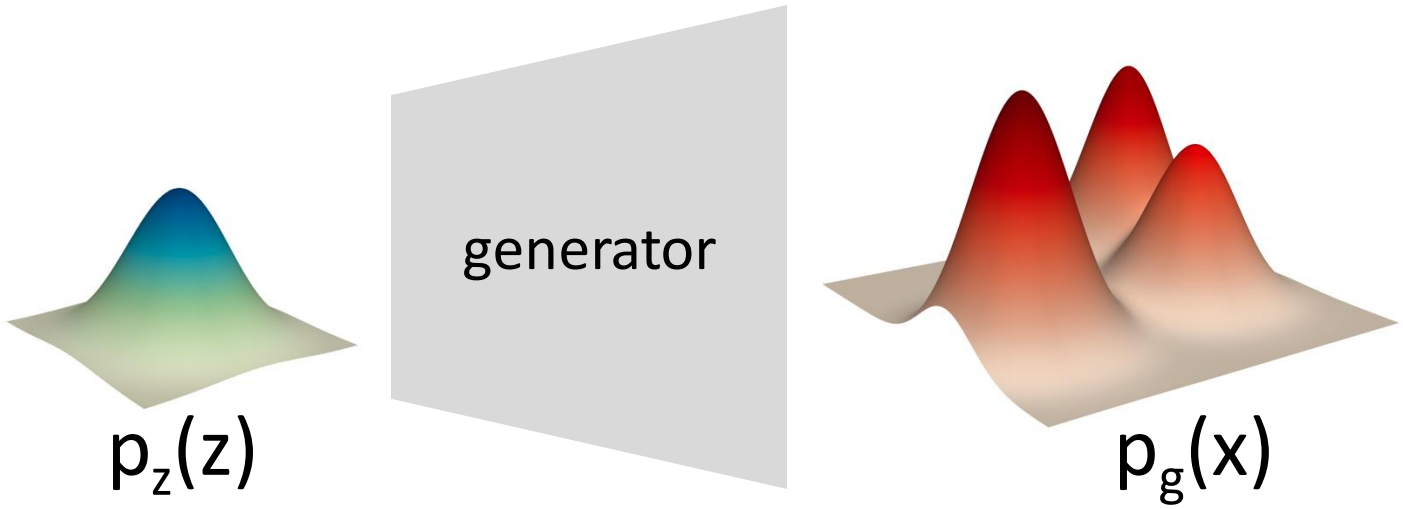
- Generative modeling is a way of problem-solving.

# Families of Generative Models

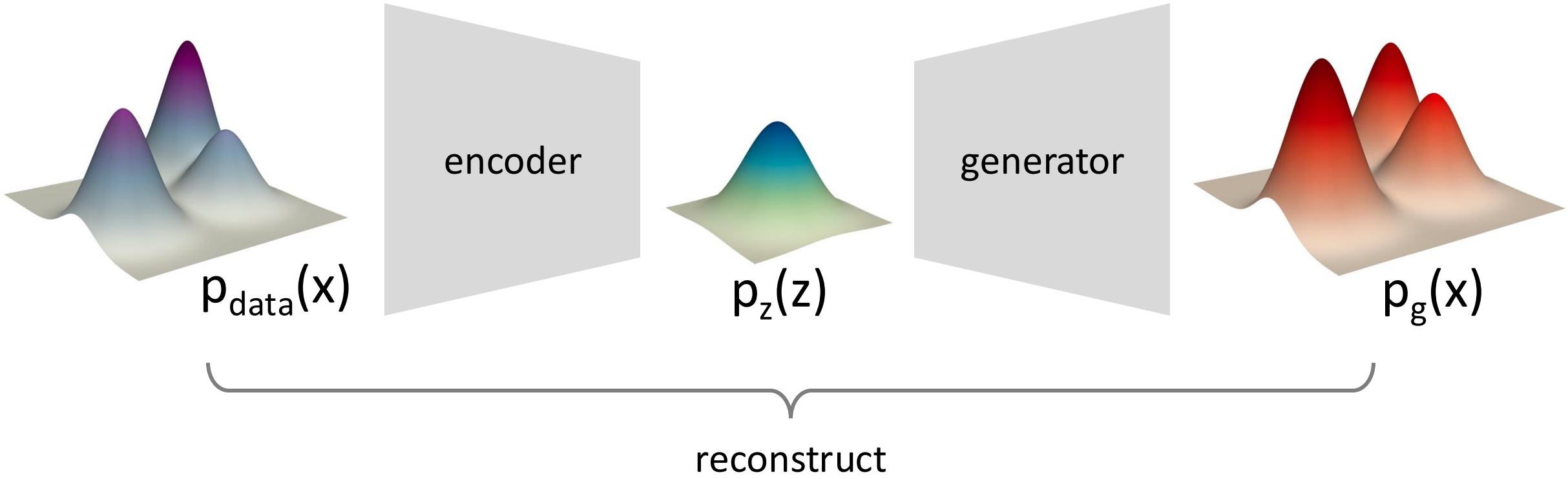
# [Recap] Learning to Represent Distributions



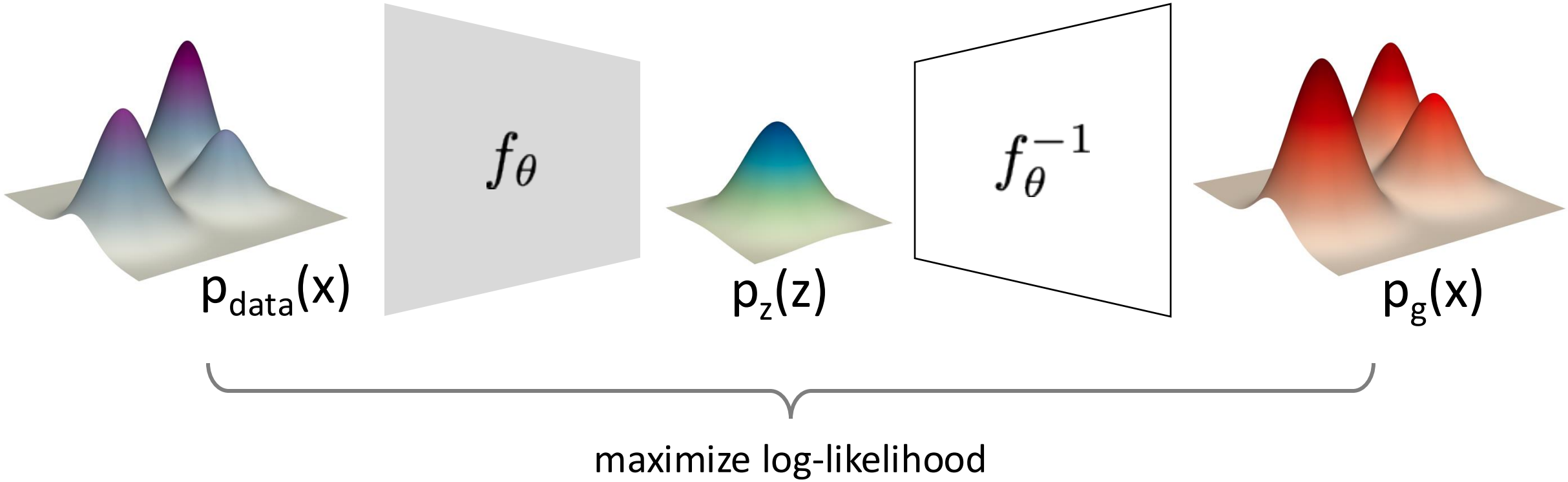
# [Recap] Learning to Represent Distributions



# Variational Autoencoder (VAE)

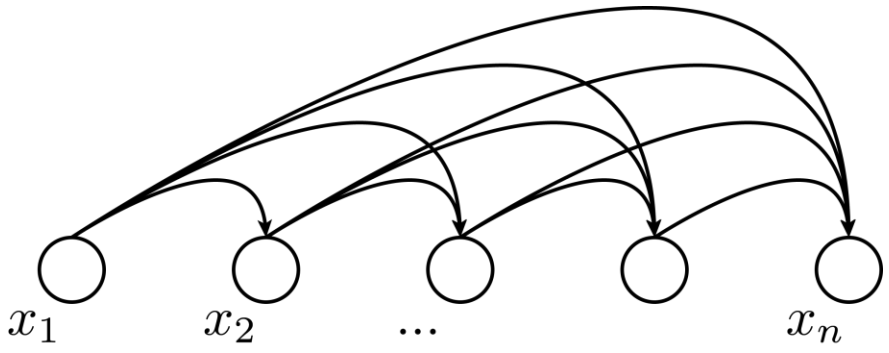


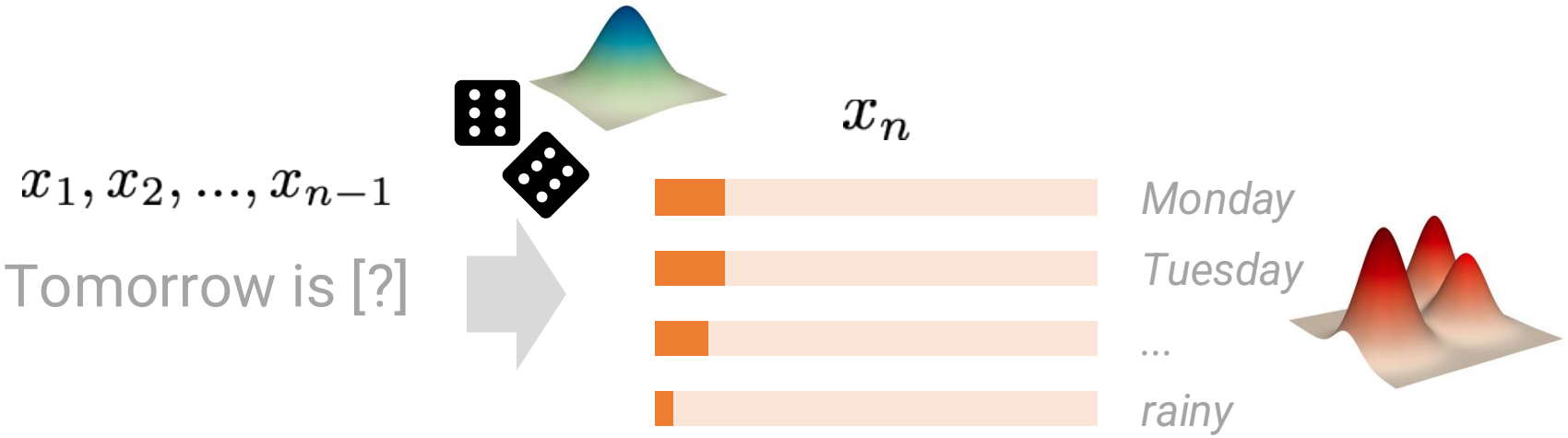
# Normalizing Flows



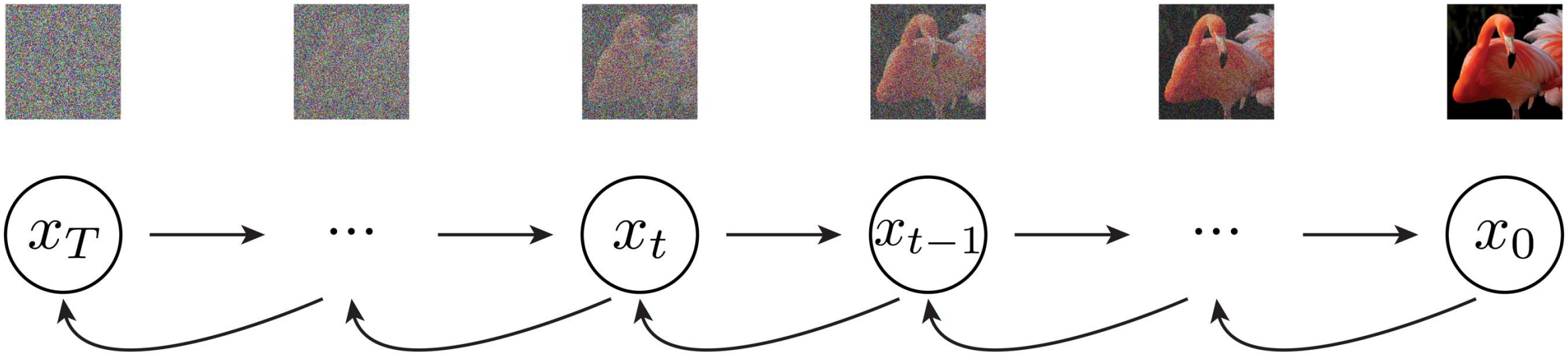
# Autoregressive Models (AR)

$$p(x_1, x_2, ..., x_n) = p(x_1)p(x_2 \mid x_1)...p(x_n \mid x_1, x_2, ..., x_{n-1})$$

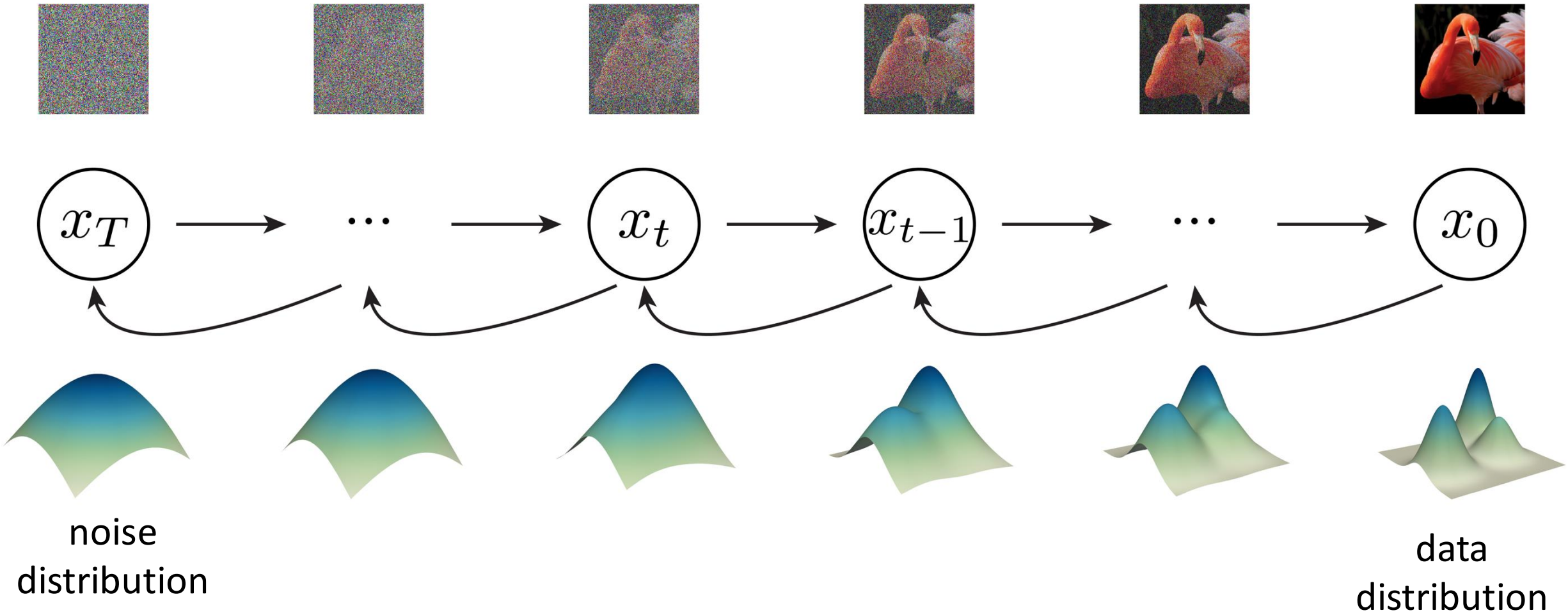


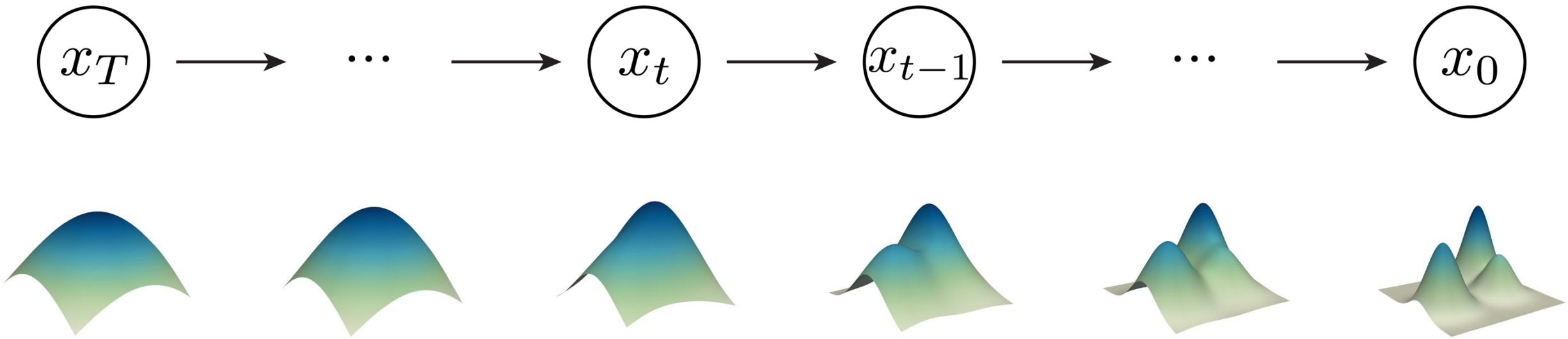
# Diffusion Models



# Diffusion Models



# Diffusion Models



# Flow Matching

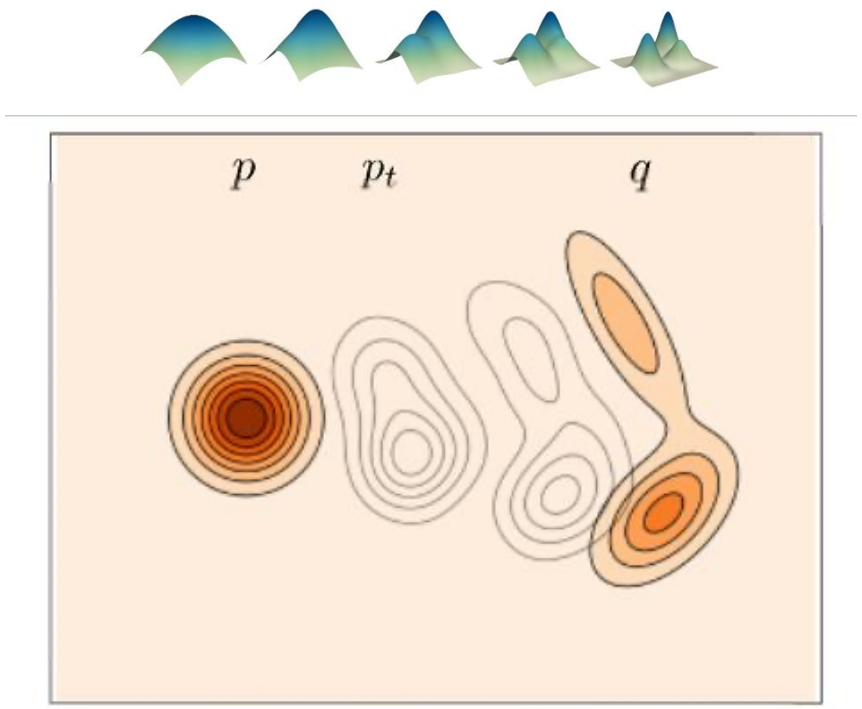
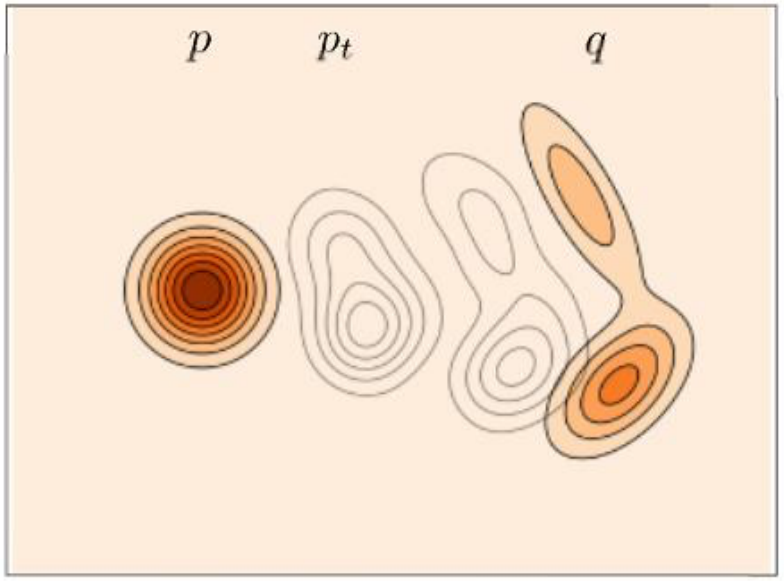
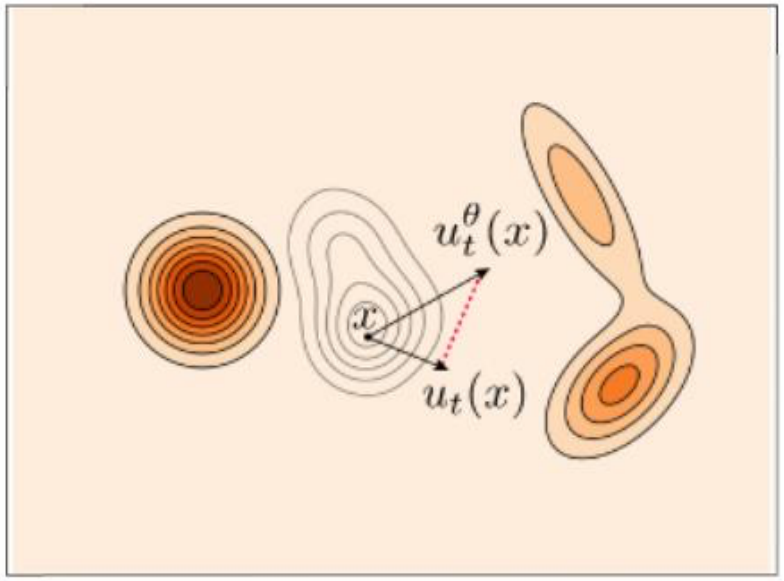



Figure adapted from: https://github.com/facebookresearch/flow_matching

# Flow Matching



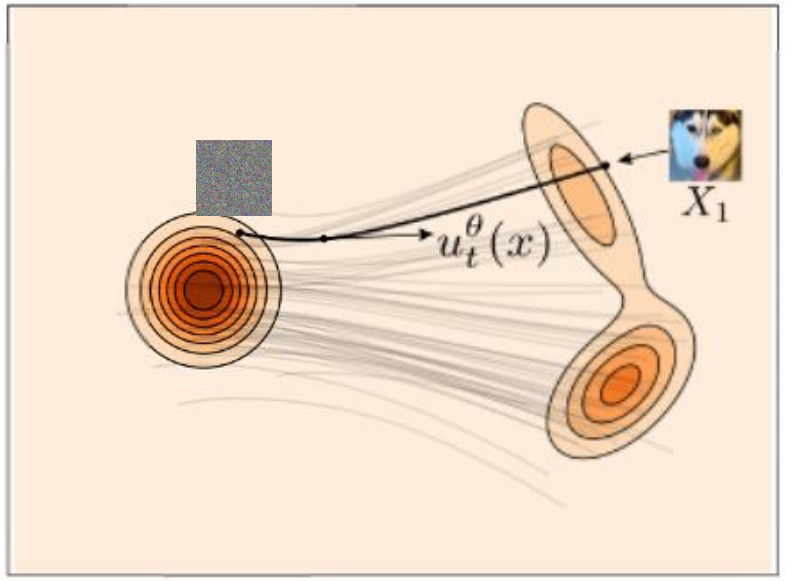




Figure adapted from: https://github.com/facebookresearch/flow_matching

# Takeaways

- Generative modeling is probabilistic modeling
- Learning to represent probabilistic distributions
- A way of problem-solving